

Gourab Roy

AI Researcher & Erasmus Mundus IPCVAI Student

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Research Interests

Computer vision, deep learning, explainable AI (XAI), vision transformers, CLIP-style vision-language models, biomedical imaging (CT/MRI), few-shot learning, and trustworthy AI for real-world deployment.

Education

Erasmus Mundus Joint Master's Degree — IPCVAI

2025–2027 (ongoing)

Artificial Intelligence for Image Processing and Computer Vision

- **Semester 2 (current):** Universidad Autónoma de Madrid (UAM), Spain *Feb 2026 – Present*
Courses: HPC for Deep Learning, DL for Visual Signal Processing I, DL for Computer Vision II, Vision for Multiple/Moving Cameras, Tutored Research & Development (TRDP 1)
- **Semester 1 (completed):** Pázmány Péter Catholic University (PPKE), Budapest, Hungary *Sep 2025 – Jan 2026*
Courses: Basic Image Processing, Data Mining & ML, Fundamentals of Deep Learning, Multimodal Sensor Fusion & Navigation, Biomedical Signal Processing, Design Patterns
- **Semester 3 (planned):** Université de Bordeaux (UBx), France
Courses: Explainable AI for Images (XAI), AI & Sustainability, Advanced Computer Vision, Acquisition, Reconstruction & Medical Imaging

B.Sc. Computer Science and Engineering

2018–2023

Sylhet Engineering College, Bangladesh

CGPA: 3.56/4.00

- Undergraduate thesis: *CerebralNet* (deep learning applied to brain signal/image analysis)

Experience

Research Assistant

Nov 2023 – Present

Sylhet Engineering College, Department of CSE

Supervisor: Prof. Md. Abu Naser Mojumder, Head of CSE

- Led applied ML research: model design, experimental evaluation, and paper preparation across medical AI, document AI, NLP, and structural CV — contributing to 8 published studies
- Mentored undergraduate students on final-year thesis and research projects

Adjunct Lecturer

Jan 2024 – Present

Sylhet Engineering College, Department of CSE

- Taught undergraduate courses: Distributed Computing, Computer Architecture, Introduction to Data Science, Peripheral & Interfacing, Microprocessors & Microcontrollers
- Designed assessments, held consultation sessions, and co-supervised final-year thesis projects

Publications — [Google Scholar](#) · 24+ citations

Journal Articles

- [J1] “CovidExpert: A Triplet Siamese Neural Network framework for the detection of COVID-19,” *Informatics in Medicine Unlocked*, Elsevier, 2023. [doi:10.1016/j.imu.2022.101156](https://doi.org/10.1016/j.imu.2022.101156) [22 citations]
- [J2] “Optical character recognition for ancient scripts: a case study on Syloti Nagri using deep learning models,” *Journal of Computer Sciences Institute*, 2025. [\[link\]](#)
- [J3] “Advancing Bangla typography: machine learning and transfer learning based font detection and classification using the BanglaFont45 dataset,” *Journal of Computer Sciences Institute*, 2025. [\[link\]](#)

Conference Papers

- [C1] “CrackClassification4: Classifying Four Types of Cracks in Structural Surfaces Using CNNs,” *ICCIT 2024*, IEEE. [\[IEEE Xplore\]](#)

- [C2] “Optimizing Multilingual Summarization: Advanced Corpus Creation and Filtering Approach,” *ICCIT 2024*, IEEE. [1 citation]
- [C3] “Enhancing Concrete Performance: Machine Learning Models for Predicting and Classifying Tensile Strength,” *ECCE 2025*, IEEE. [IEEE Xplore]
- [C4] “DateFNet: From Pixels to Plates — Attention Based Multi-Stream CNN for Date Fruit Classification,” *ICCIT 2025*, IEEE.
- [C5] “Deep Learning-Based Detection of Aircraft Cracks and Dents Using YOLOv11 and YOLOv12 for Aviation Safety in Bangladesh,” *ICCIT 2025*, IEEE.

Selected Projects

Cross-Modal Attention Bridge (CMAB) for CLIP Interpretability UAM Madrid, Feb 2026 – Ongoing

- Developing a statistically filtered cross-modal attention method explaining CLIP’s image–text alignment; K-sigma filtering on attention rollouts generates fine-grained, text-conditional visual explanations per token without altering the model weights
- Supervised by Prof. Jenny Benois-Pineau (UBx Bordeaux) and Prof. Marcos Escudero-Viñolo (UAM Madrid) — *TRDP 1 research project*
- *Tools*: PyTorch · ViT · CLIP · Attention Rollout · XAI

Cardiac Autonomic Assessment via PPG-Derived HRV During Sound Stimulation PPKE Budapest, Dec 2025

- Modular Python pipeline: 10 participants across baseline and music conditions; Butterworth filtering, adaptive peak detection, FFT spectral analysis, Friedman/Wilcoxon tests and individual responsiveness classifier — every subject showed significant ANS shifts despite non-significant group means
- github.com/GourabRoy551/cardiac-autonomic-ppg-hrv
- *Tools*: Python · NumPy · SciPy · Pandas · Matplotlib

Calibrated Multi-View 3D Reconstruction UAM Madrid, 2025

- Full MATLAB pipeline: camera calibration, SIFT/SURF/ORB feature matching with RANSAC, fundamental matrix, epipolar geometry, multi-view sparse reconstruction, and bundle adjustment — 4.47 px reprojection error over 672 multi-view tracks
- github.com/GourabRoy551/Calibrated-Multi-View-3D-Reconstruction
- *Tools*: MATLAB · SIFT · RANSAC · Bundle Adjustment · Epipolar Geometry

Technical Skills

Languages	Python · C · C++ · Java · JavaScript
Deep Learning	PyTorch · TensorFlow · Scikit-learn · CNNs · Vision Transformers (ViT) · YOLO (v11/v12) · CLIP · Attention Rollout · Transfer learning · Few-shot learning
Computer Vision	Image classification · Segmentation · Object detection · Multi-view geometry · Explainable AI (XAI)
Data Analysis	NumPy · Pandas · Matplotlib · SciPy · Jupyter Notebook
Dev Tools	Git · MATLAB · MySQL · SQLite
Platforms	Linux · Windows
Research	L ^A T _E X · Academic writing · Peer review (<i>Informatics in Medicine Unlocked</i> , Elsevier)

Academic Service & Achievements

- **Peer Reviewer**, *Informatics in Medicine Unlocked* (Elsevier)
- **HackerRank Gold Badges** in C++ and Python
- **350+** competitive programming problems solved (HackerRank, Codeforces, UVA)
- Completed Machine Learning Specialization and data science courses (Coursera)